

FINITE FAILURE SCHEMES AND THE RECONSTRUCTION OF QUADRATIC PENCILS

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ABSTRACT. We reconstruct every complex quadratic pencil from an abstract unmarked finite-order neighbourhood of its multiplication-failure scheme. For embedding dimension $n \geq 3$, the smallest uniform order is $d = n^2 + 2n - 4$; every lower-order neighbourhood is independent of the pencil. The inverse recovers a tensor orientation from intrinsic rank-one rulings and then recovers the full coefficient line. In fact, a neighbourhood supported on a single Schubert line suffices. More generally, the same reconstruction works on any smooth projective curve with a rank- n bundle having nontrivial projectivization. We construct the actual universal finite neighbourhood, compatible with arbitrary complex base change with its relative socle retained. For regular pencils its spectral readout includes all elementary divisors and higher Fitting incidence schemes. Fixed-discriminant specializations illustrate the distinction between this information and reduced rank data. The unmarked inverse, the framed parameter immersion, and the relative constructions are kept separate.

1. INTRODUCTION

A degeneracy locus usually forgets the map that defines it. We study an inverse problem in which neither the ambient embedding nor the normal directions are retained: can the abstract nonreduced failure scheme recover a quadratic pencil? The answer is affirmative for every complex pencil, including singular pencils. More precisely, there is a smallest uniform infinitesimal order at which the pencil appears. The global geometry needed for this inverse is small: one projective line in the socle already suffices.

Let V be a complex vector space of dimension $n \geq 3$, and put

$$W = \mathrm{Sym}^2 V, \quad N = \binom{n+1}{2}, \quad R \in \mathrm{Gr}(2, W), \quad S_R = W/R, \quad p = N - 2.$$

The cube-zero algebra $A_R = \mathbb{C} \oplus V \oplus S_R$ has multiplication $\mathrm{Sym}^2 V \rightarrow S_R$, with $(V \oplus S_R)S_R = 0$. On $X_R = \mathrm{Gr}(n, V \oplus S_R)$, let $\mathcal{K}$ denote the tautological bundle. For any $m \geq 2$, set

$$\widehat{D}_R = V(\mathrm{Fitt}_0 \mathrm{coker}[\mathrm{Sym}^m(\mathcal{O} \oplus \mathcal{K}) \rightarrow A_R \otimes \mathcal{O}]). \quad (1.1)$$

This is the maximal-minor scheme of multiplication, not merely its support. It is independent of $m \geq 2$. Its reduction is the Schubert divisor where $\mathcal{K} \rightarrow V$ drops rank. Write $B_R = \mathrm{Gr}(n, S_R)$, let $\mathfrak{b}_R$ be its ideal in $\widehat{D}_R$, and define

$$Z_R^{[k]} = V(\mathfrak{b}_R^{k+1}) \subset \widehat{D}_R, \quad d = n + 2p = n^2 + 2n - 4.$$

An isomorphism of these neighbourhoods means an abstract complex scheme isomorphism; it is supplied with no ambient or tensor marking.

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Theorem 1.1 (Sharp intrinsic reconstruction). *For all quadratic pencils $R, R' \subset \text{Sym}^2 V$,*

$$Z_R^{[d]} \simeq Z_{R'}^{[d]} \iff R' = gR \text{ for some } g \in \text{PGL}(V).$$

For every $0 \leq k < d$, the isomorphism class of $Z_R^{[k]}$ is independent of R . Thus d is the smallest uniform reconstruction order. The same statements hold after restriction, using the graph projection, to any Schubert line in B_R . The resulting abstract curve-supported neighbourhood is independent of the choice of that line.

The Grassmannian statement is Theorem 5.1. The curve-supported statement is Corollary 6.2. Its more general form, Theorem 6.1, applies over any smooth connected projective curve with a rank- n bundle having nontrivial projectivization. For example, $\mathcal{O}_C(-P) \oplus \mathcal{O}_C^{n-1}$ works on every such pointed curve. The curve is intrinsic as the reduction; a chosen embedding of the curve is not part of the isomorphism data.

1.1. The intrinsic mechanism. In the graph neighbourhood of B_R , with tautological bundle $\mathcal{U}$ and graph map $T : \mathcal{U} \rightarrow V$, the ideal is

$$(\det T)I_p(\gamma_R \text{Sym}^2 T).$$

Its generators have degree d . This degree calculation alone only explains why lower orders are constant. The inverse requires three further intrinsic operations. Multiplication in the finite scheme recovers the first relation kernel and hence the whole homogeneous cone. The rank-one Fano scheme of its reduced determinant cone then recovers the two tensor rulings. Precisely one is projectively trivial over the reduction; the other is distinguished already on a Schubert line. Properness forces the induced map to $\text{PGL}(V)$ to be constant. Finally, cancellation of the intrinsic determinant ideal recovers the coefficient module. For pencils its exterior representation is irreducible, and exterior duality identifies its single coefficient line with the Pluecker line of R .

Theorem 3.1 separates this geometric mechanism from the particular multiplication algebra. The curve theorem shows why the mechanism is not tied to the dimension of the original parameter Grassmannian. A single unmarked Artinian normal slice does not supply the global ruling distinction used in the proof; we make no corresponding local-algebra classification assertion.

1.2. Framed families and spectral consequences. Once tensor coordinates are retained, the parameter map is simpler. The first relation spaces give a closed immersion of $\text{Gr}(2, W)$ into a Grassmannian, by Theorem 5.3. Its proof factors through the classical Pluecker embedding, tensoring a line with a fixed space, and a fixed linear inclusion. This is the framed encoding consequence of the coefficient calculation, not a substitute for the unmarked inverse. Theorem 8.1 glues the actual universal order- d neighbourhood. It is finite locally free over the relative socle of rank $\binom{n^2+d}{d} - \binom{N}{2}$, and projective and flat over the pencil parameter scheme. It commutes with arbitrary complex base change with that socle retained. Over a nonreduced base, the specified relative socle ideal need not be the absolute nilradical. These statements do not identify unmarked moduli stacks or deformation groupoids.

For regular pencils, the spectral sheaf and its Fitting incidence schemes are read from the reconstructed pencil. We retain these consequences because they exhibit information genuinely lost by coarser invariants: fixed discriminants and fixed reduced rank loci can conceal different elementary divisors. In Theorem 9.2, every classical dominance specialization is transported to a flat family of the finite neighbourhoods, with every strict transition detected at order d . The spectral orbit classification itself is classical, not a second independent Torelli theorem.

1.3. Prior work and the publication object. The Weierstrass–Segre classification is used in its root-labelled form; compare [4, Theorem 1.1 and Corollary 2.1]. The orthogonal nilpotent input is isolated in Lemma 9.1, with precise references to [5, Proposition 1 and Theorem 1], an independent dimension calculation, and an explicit distinction between full orthogonal orbits and their special-orthogonal components.

Ballico’s relative multiplication construction [2, Theorem 0.2; §1, Remark 1.2 and equation (6)] is a substantive antecedent for studying quadratic-normality failure on finite linear sections of an embedded surface. Here the defining algebra varies, and the inverse starts from an unmarked infinitesimal failure scheme. A theorem-level comparison with the distinct earlier paper [1] still requires its complete text; no claim of nonanticipation relative to that unread text is made.

The present article is the reconstruction argument and its spectral readout. The reciprocal-likelihood theorem and projective critical schemes are preserved in a separate applications manuscript. Its finite-flat critical divisor requires supplied split semisimple data and simple disjoint projective poles; its totally real cover requires a supplied real definite realization and separated data. None of those hypotheses enters the all-pencil theorem here. Independent web, exterior-operator, K3, polarized, and boundary results are preserved in an explicitly non-submitted research archive. They are not inputs to any proof in this article and are not part of its significance claim. Every proof required here is included here.

2. INTRINSIC TENSOR RULINGS

The first step of the inverse problem is to recognize the two tensor factors of a determinant cone without an ambient marking. The following scheme-theoretic ruling calculation and descent lemma supply precisely this step. They apply to the pencil argument independently of the web and exterior-contraction constructions in the separate research archive.

Lemma 2.1 (The relative scheme of maximal rank-one spaces). *Let B be any complex scheme and E, F vector bundles of rank $n \geq 2$. In $\mathrm{Gr}_B(n, E \otimes F)$, let $Z(E, F)$ be the zero scheme of the restriction of the quadratic rank-one equations to the universal subbundle $\mathcal{T}$. Equivalently it is the zero scheme of*

$$\mathrm{Sym}^2 \mathcal{T} \longrightarrow \bigwedge^2 E \otimes \bigwedge^2 F.$$

The natural morphism

$$\mathbb{P}_B(E) \sqcup \mathbb{P}_B(F) \longrightarrow Z(E, F), \quad \ell \mapsto \ell \otimes F, \quad m \mapsto E \otimes m, \quad (2.1)$$

is an isomorphism. It commutes with arbitrary change of complex base, including change to a nonreduced base.

Proof. We first work over $\mathbb{C}$. Each displayed map to the Grassmannian is a closed immersion. Its composition with the Pluecker embedding is the degree- n Veronese embedding on $\ell^{\otimes n} \otimes \det F$, respectively $\det E \otimes m^{\otimes n}$, followed by its natural linear inclusion into $\bigwedge^n(E \otimes F)$. This linear inclusion is injective: polarization of the nonzero pure-power map gives the corresponding irreducible symmetric-power summand. The rank-one equations vanish identically on both universal families, so the closed immersions factor through $Z(E, F)$.

Every geometric point is in one of the two images. Indeed the sum of two rank-one tensors is rank at most one only if their first factors or their second factors are proportional. Starting with two independent tensors, this observation forces a linear space consisting of rank-one tensors to fix the same factor throughout. A space of dimension n is then exactly one of the displayed spaces. The two images are disjoint: a space fixing both factors has dimension one, not n .

We verify that these pointwise statements omit no infinitesimal structure. At $\ell \otimes F$, choose a generator of ℓ and a complement in E . A Grassmannian tangent vector is a map $\phi : F \rightarrow (E/\ell) \otimes F$.

For each coordinate of E/ℓ , write its component as $A : F \rightarrow F$. Linearizing the restricted minors gives

$$u \wedge A(u) = 0 \quad \text{for every } u \in F.$$

Polarizing this diagonal identity gives

$$u \wedge A(v) + v \wedge A(u) = 0 \quad (u, v \in F).$$

In characteristic zero the diagonal and polarized identities are equivalent. These are precisely all coefficients of the bilinear linearization of the restricted quadratic map on $\text{Sym}^2 F$, so no infinitesimal equation is omitted by testing the diagonal. Taking the basis vectors shows that A is diagonal; taking their pairwise sums makes all its diagonal entries equal. Conversely a scalar A satisfies all the linearized equations. Thus the Zariski tangent space to $Z(E, F)$ is precisely $\text{Hom}(\ell, E/\ell)$, of dimension $n - 1$. The other image has the symmetric calculation.

At every closed point the closed immersed projective space gives local dimension at least $n - 1$, whereas the tangent calculation gives embedding dimension exactly $n - 1$. The local ring is therefore regular. A finite-type complex scheme whose closed-point local rings are regular is reduced: a nonempty support of its nilradical would contain a closed point. In particular there is no nilpotent thickening or embedded associated structure at either image. The disjoint union of the two closed immersions is itself a closed immersion, surjective on geometric points. Its ideal in the now reduced target vanishes at every closed point, so it is zero. This proves the scheme isomorphism over $\mathbb{C}$.

For general B , trivialize E, F on a common open cover. The Grassmannian and the zero scheme of the displayed universal quadratic map are, on such an open, the base change of the constant construction just proved. Thus (2.1) is an isomorphism there even if the base has nilpotents. The maps are natural under changes of frame and hence glue. This argument is an identification by equations and base change, not an inference of relative flatness from a fibrewise tangent calculation. $\square$

Lemma 2.2 (Oriented relative Segre descent). *Let B be a connected projective complex variety with $H^0(B, \mathcal{O}_B) = \mathbb{C}$, and let V, V' have the same dimension $n \geq 2$. Let $\mathcal{U}, \mathcal{U}'$ be vector bundles of rank n on B . Suppose a linear bundle isomorphism*

$$\alpha : V \otimes \mathcal{U}^* \xrightarrow{\sim} V' \otimes \mathcal{U}'^*$$

preserves the rank-one cones and the ruling whose parameter bundle is $\mathbb{P}(V) \times B$. The induced isomorphism $\mathbb{P}(V) \times B \simeq \mathbb{P}(V') \times B$ is regular and is induced by one constant projective isomorphism $[g] : \mathbb{P}(V) \rightarrow \mathbb{P}(V')$. Locally $\alpha(T) = gTh^{-1}$, with the same $[g]$ on every chart. The possible line-bundle twists of vector-bundle lifts do not change the projective left coefficient lines of any homogeneous functorial coefficient construction.

Proof. Lemma 2.1, applied to $E = V \otimes \mathcal{O}_B$ and $F = \mathcal{U}^*$, identifies the closed relative Fano scheme with $(\mathbb{P}(V) \times B) \sqcup \mathbb{P}_B(\mathcal{U}^*)$. In particular its two components carry their displayed scheme structures, with no additional nilpotent or embedded components. The isomorphism α induces a regular map of these relative parameter schemes, and the orientation hypothesis preserves the first component.

For completeness, an isomorphism of two trivial $\mathbb{P}^{n-1}$ -bundles over any base gives a regular projective-matrix-valued map, not merely a fibrewise collection. Its pullback of the relative $\mathcal{O}(1)$ has the form $\mathcal{O}(1) \otimes p^*L$ for a line bundle L on the base. To see this, tensor by $\mathcal{O}(-1)$; the resulting line bundle is trivial on every projective-space fibre, and its pushforward and evaluation identify it with a pullback from the base. Pushing forward $\mathcal{O}(1)$ now gives an isomorphism of the corresponding trivial rank- n bundles, with that line-bundle twist. On charts trivializing L , it is an invertible matrix g_i of regular functions. On overlaps, $g_j = u_{ij}g_i$ for an invertible regular

function u_{ij} . The classes $[g_i]$ thus glue to a regular morphism to the projective-isomorphism scheme, identified after choosing fixed bases with PGL_n . This accounts explicitly for the possible absence of a global linear lift.

The group PGL_n is affine: it is the determinant-nonzero principal open in the projective space of matrices. A morphism from B to an affine scheme factors through $\mathrm{Spec} H^0(B, \mathcal{O}_B) = \mathrm{Spec} \mathbb{C}$, so the projective map is constant. Choose a constant linear lift g . Each local g_i then differs from g by a scalar unit. After removing its two projective-factor actions, a linear map preserving every decomposable line is scalar: apply it to sums of two tensors with one common factor. Consequently the remaining action is a right bundle map, and the scalar units may be absorbed in its local lifts h . This proves the local tensor formula.

Finally a scalar multiplying a lift acts on a homogeneous functor of degree a by its scalar to the power a (or $-a$ on its dual). It cannot alter a projective coefficient line. The conclusion applies to every homogeneous degree. In particular, no right change of frame can replace the single projective left map by separate transformations on different coefficient spaces. $\square$

3. AN INTRINSIC COEFFICIENT PRINCIPLE

Extracting coefficient data from an abstract scheme and reconstructing a defining relation space from those data are distinct operations. The criterion below isolates the first. Its hypotheses will be verified directly for the multiplication schemes and then on curves.

3.1. The structural criterion. Let X be a finite-type complex scheme with a closed subvariety B specified by a construction preserved by every abstract scheme isomorphism. Write $\mathfrak{b} \subset \mathcal{O}_X$ for its ideal and

$$\mathcal{G} = \bigoplus_{r \geq 0} \mathfrak{b}^r / \mathfrak{b}^{r+1}, \quad \mathcal{A} = \mathrm{Sym}_{\mathcal{O}_B}(\mathcal{G}_1), \quad \mathcal{I} = \ker(\mathcal{A} \twoheadrightarrow \mathcal{G}).$$

Assume the following conditions, for fixed integers $n \geq 2, m \geq 1$.

- (i) B is connected and projective, $H^0(B, \mathcal{O}_B) = \mathbb{C}$, and $\mathcal{G}_1$ is locally free. Its dual admits a presentation $V \otimes \mathcal{U}^*$, where $\dim V = \mathrm{rank} \mathcal{U} = n$.
- (ii) The reduction of $\mathrm{Spec}_B \mathcal{G}$ in its degree-one ambient bundle is the generic determinant cone. The bundle $\mathbb{P}_B(\mathcal{U}^*)$ is not projectively trivial. Thus precisely one of the two ruling parameter bundles of its rank-one locus is projectively trivial.
- (iii) If $\mathcal{D} \subset \mathcal{A}$ is the ideal of that reduced cone and $\mathcal{L} = \mathcal{D}_n$, then $\mathcal{I} = \mathcal{D}\mathcal{J}$, where $\mathcal{J}$ is generated in degree m . Under the Cauchy decomposition in any local frame of $\mathcal{U}$,

$$\mathcal{J}_m = \bigoplus_{\lambda \in \Lambda} L_\lambda \otimes \mathbb{S}_\lambda \mathcal{U} \subset \mathrm{Sym}^m(V^* \otimes \mathcal{U}), \quad L_\lambda \subset \mathbb{S}_\lambda V^*, \quad (3.1)$$

where Λ is a specified collection of partitions of m of length at most n , and the left subspaces are constant over B . Zero subspaces are permitted. The Cauchy inclusions, not just their representation labels, are the coefficient realizations $\beta \otimes \xi \mapsto [T \mapsto \beta(\mathbb{S}_\lambda(T)\xi)]$.

These conditions are hypotheses on the normal cone and its residual coefficient module. They do not assume that an abstract isomorphism extends to an ambient space, preserves a chosen frame, or already acts on all the left subspaces through a common projective transformation.

Theorem 3.1 (Intrinsic coefficient reconstruction). *Suppose X, B satisfy (i)–(iii). The abstract scheme X determines the collection $(L_\lambda)_{\lambda \in \Lambda}$ up to one common projective coordinate transformation on V . More precisely, an isomorphism between two such schemes induces a constant projective isomorphism $[g] : \mathbb{P}(V) \rightarrow \mathbb{P}(V')$ such that*

$$L_\lambda = \mathbb{S}_\lambda(g)^* L'_\lambda \quad (\lambda \in \Lambda).$$

The equality means equality of subspaces; scalar changes of a linear lift of $[g]$ do not affect it.

Proof. An isomorphism transports B , every power of $\mathfrak{b}$, $\mathcal{G}_1$, and the surjection $\mathcal{A} \twoheadrightarrow \mathcal{G}$. In particular it transports $\mathcal{I}$ in the intrinsic degree-one ambient bundle. Iterated reduced singular loci of the determinant cone give its rank-one locus. Lemma 2.1 reconstructs its two ruling parameter schemes, including their scheme structure. Condition (ii) distinguishes the trivial ruling intrinsically. Lemma 2.2 then gives one constant projective map $[g]$. After choosing a constant linear lift and local frames on both bases, the ambient bundle map has the form $T \mapsto gTh^{-1}$. These local choices are made only after constructing the graded algebra, its degree-one part, and the ruling.

Locally $\mathcal{D} = (d)$ with $d = \det T$. Its leading coefficient is a unit, so it is a nonzerodivisor over every coefficient ring. Hence $(\mathcal{I} : \mathcal{D}) = \mathcal{J}$, and

$$\mathcal{J}_m \simeq \mathcal{L}^* \otimes \mathcal{I}_{n+m}.$$

This recovers the residual module without choosing a scalar determinant equation. Under $T \mapsto gTh^{-1}$, the Cauchy coefficient map transports a tensor by $\mathbb{S}_\lambda(g)^* \otimes \mathbb{S}_\lambda(h^{-1})$. Projection to the specified Cauchy summand, followed by contraction against its whole right dual, therefore transports exactly L'_λ to L_λ . The right map is invertible and has no effect on that left image. Both determinant-line transition functions and changes of a lift of $[g]$ act by scalar units. Since every partition has size m , this ambiguity is common in degree and disappears on subspaces. This proves the assertion. $\square$

4. QUADRATIC RELATION SPACES AND INTRINSIC RECONSTRUCTION OF PENCILS

The multiplication construction itself supplies a wider natural class for the coefficient criterion. In this section the number of quadratic relations is allowed to vary independently of the embedding dimension. For two relations the resulting theorem recovers every pencil of quadrics, including singular pencils. No extra block is added to the multiplication map.

4.1. Automatic coefficient extraction for quadratic multiplication. Let $\dim V = n \geq 3$, $W = \text{Sym}^2 V$, $N = n(n+1)/2$, and $R \in \text{Gr}(r, W)$, where $1 \leq r < N - n$. Put $p = N - r > n$, $S = W/R$, and

$$A_R = \mathbb{C} \oplus V \oplus S, \quad X_R = \text{Gr}(n, V \oplus S).$$

The only nonzero product in the maximal ideal is the quotient map $\gamma : \text{Sym}^2 V \rightarrow S$. Write $\mathcal{K}$ for the tautological bundle on X_R , and use the same definition (1.1) of $\widehat{D}_R$, with A_R in place of B_R . Thus $\widehat{D}_R$ is the failure scheme of generating A_R by a varying n -plane and the unit. The notation in this section agrees with the web construction when $(n, r) = (4, 4)$.

Let $\Lambda_{n,p}$ denote the partitions occurring in the classical multiplicity-free decomposition

$$F(V) := \bigwedge^p \text{Sym}^2 V = \bigoplus_{\lambda \in \Lambda_{n,p}} \mathbb{S}_\lambda V. \quad (4.1)$$

Every partition has size $2p$; terms of length greater than n are omitted. Choose a nonzero volume form on S and let $\beta_R \in F(V)^*$ be its pullback by $\bigwedge^p \gamma$. Its component in $\mathbb{S}_\lambda V^*$ is denoted by $\beta_{R,\lambda}$. Only the lines of the nonzero components will be used, so the volume choice is immaterial.

Theorem 4.1 (Automatic intrinsic coefficient extraction). *For every $R \in \text{Gr}(r, \text{Sym}^2 V)$ in the stated range, the abstract scheme $\widehat{D}_R$ determines*

$$(\mathbb{C}\beta_{R,\lambda})_{\lambda \in \Lambda_{n,p}}$$

up to one common projective transformation of V , with zero components recorded as zero subspaces. More precisely, the intrinsic construction recovers the base $B = \text{Gr}(n, S)$, its degree-one normal-cone bundle, and the residual ideal generated in degree $2p$ with coefficient module

$$\mathcal{J}_{2p} = \bigoplus_{\lambda \in \Lambda_{n,p}} \mathbb{C} \beta_{R,\lambda} \otimes \mathbb{S}_\lambda \mathcal{U} \subset \text{Sym}^{2p}(V^* \otimes \mathcal{U}), \quad (4.2)$$

where $\mathcal{U}$ is the tautological rank- n bundle on B . Consequently all geometric hypotheses of Theorem 3.1 hold for this entire Grassmannian family, not just for smooth webs.

Proof. On a frame chart of $\mathcal{K}$, write its inclusion as $\begin{pmatrix} M \\ A_0 \end{pmatrix}$, with M an $n \times n$ matrix. Since A_R is cube-zero, columns of degree greater than two vanish. After retaining the scalar, linear, and quadratic columns, multiplication has the matrix

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & M & 0 \\ 0 & A_0 & \gamma \text{Sym}^2 M \end{pmatrix}. \quad (4.3)$$

A nonzero maximal minor must use the scalar column and all n linear columns: these are the only columns meeting the first $1+n$ rows. Expanding along those rows gives the identity of ideals

$$I_{\widehat{D}_R} = (\det M) I_p(\gamma \text{Sym}^2 M). \quad (4.4)$$

The same expansion works for every $m \geq 2$. Where M is invertible, the quadratic map is surjective. Hence the reduction of $\widehat{D}_R$ is exactly the Schubert divisor where M is singular, independent of R after identifying the vector spaces S .

Successive reduced singular loci of that divisor are the projection rank loci. Their last nonempty member is $B = \text{Gr}(n, S)$. For clarity, the usual determinantal assertion here is local: at a projection-rank- k point, an invertible k -minor splits away, leaving a generic square $(n-k)$ -matrix and smooth coordinates. Every remaining entry can vary independently by moving the kernel of the projection in V modulo its image. Thus its reduced singular locus is the next rank locus. Since $p > n$, the final base is nonempty, connected and projective, with $H^0(B, \mathcal{O}_B) = \mathbb{C}$.

Near B , a Grassmannian chart consists of a varying plane in S and its graph $T : \mathcal{U} \rightarrow V$. Formula (4.4) is independent of the other graph coordinates and homogeneous in T . It therefore gives its actual normal-cone ideal, not merely an initial reduced support:

$$C_B \widehat{D}_R \subset \text{Tot Hom}(\mathcal{U}, V), \quad \mathcal{I} = (\det T) \mathcal{J}, \quad \mathcal{J} = I_p(\gamma \text{Sym}^2 T). \quad (4.5)$$

There are no degree-one equations. The ambient bundle in (4.5) is consequently the dual of the intrinsically recovered degree-one conormal. Its reduced ideal is the determinant ideal $\mathcal{D}$. The determinant polynomial is monic for a suitable monomial order, so cancellation gives $(\mathcal{I} : \mathcal{D}) = \mathcal{J}$. This is an intrinsic ideal quotient; choosing a generator of the determinant line is not part of the recovered datum.

The rank-one locus of the reduced cone determines the two ruling parameter schemes by Lemma 2.1. They are $\mathbb{P}(V) \times B$ and $\mathbb{P}_B(\mathcal{U}^*)$. Exactly the first is projectively trivial. Indeed a Schubert line in $\text{Gr}(n, S)$ restricts $\mathcal{U}^*$ to $\mathcal{O}(1) \oplus \mathcal{O}^{n-1}$. A trivial projective bundle on that line would make this bundle a line-bundle twist of $\mathcal{O}^n$; its determinant would then have degree divisible by n , rather than degree one. Thus the trivial ruling is intrinsically oriented. Lemma 2.2 applies: an abstract isomorphism gives a morphism $B \rightarrow \text{PGL}(V)$, constant because B is connected projective and $\text{PGL}(V)$ is affine. Local linear lifts and right changes of frame give $T \mapsto gTh^{-1}$ for this one fixed $[g]$.

It remains to prove (4.2), rather than assume it as a condition on the example. The minors are the image, for fixed β_R , of the universal coefficient map

$$F(V)^* \otimes F(U) \longrightarrow \text{Sym}^{2p}(V^* \otimes U), \quad \beta \otimes \xi \longmapsto [T \mapsto \beta(F(T)\xi)]. \quad (4.6)$$

In characteristic zero, the polynomial functor F has the multiplicity-free decomposition (4.1); see [3, Theorem 3.5]. The Cauchy decomposition of the target contains each diagonal $\mathbb{S}_\lambda V^* \otimes \mathbb{S}_\lambda U$ once, and no off-diagonal pair. Equivariance kills every off-diagonal pair. On each diagonal pair the coefficient map is nonzero: identify U with V , evaluate at $T = 1$, and choose β, ξ in that summand with $\beta(\xi) \neq 0$. Schur's lemma then identifies its image with the specified Cauchy summand. This is the actual matrix-coefficient inclusion, not merely a dimension count. Keeping β_R fixed proves (4.2). All generators have degree $2p$, and at least one minor is nonzero since γ is surjective.

Under $T \mapsto gTh^{-1}$, the right map acts within the full factor $\mathbb{S}_\lambda U$. Contracting against its dual therefore leaves exactly the left coefficient line transformed by $\mathbb{S}_\lambda(g)^*$. Determinant-line scalars and local lift ambiguities do not change these lines. This proves the theorem. $\square$

The theorem does not assert that separately projectivized Schur components always determine their relative scalars. For webs this is the recombination problem solved in the separate research archive. For pencils that issue disappears altogether.

4.2. Every pencil, without a regularity hypothesis.

Lemma 4.2 (The exterior square of the quadratic representation). *For $n \geq 2$, $\bigwedge^2 \text{Sym}^2 V = \mathbb{S}_{(3,1)} V$ is irreducible. Consequently $\bigwedge^{N-2} \text{Sym}^2 V$ is irreducible as well.*

Proof. The vector $x_1^2 \wedge x_1 x_2$ is a highest-weight vector of weight $(3, 1)$. By complete reducibility it generates a copy of $\mathbb{S}_{(3,1)} V$. The hook formula gives

$$\dim \mathbb{S}_{(3,1)} V = \frac{n(n-1)(n+1)(n+2)}{8} = \binom{N}{2}.$$

This exhausts the exterior square. The last assertion follows by exterior duality. These are classical representation identities. $\square$

Theorem 4.3 (Intrinsic reconstruction of all quadratic pencils). *Let $n \geq 3$ and let R, R' be two-dimensional subspaces of $\text{Sym}^2 V$. For the natural multiplication-failure schemes of their cube-zero algebras of Hilbert function $(1, n, N-2)$,*

$$\widehat{D}_R \simeq \widehat{D}_{R'} \iff R' = gR \text{ for some } g \in \text{PGL}(V). \quad (4.7)$$

No pencil, base locus, or discriminant is required to be smooth or regular. The isomorphism on the left is abstract and carries no supplied ambient embedding, base, or tensor-factor marking. All the reduced schemes are isomorphic to the same Schubert divisor.

Proof. Here $p = N - 2 > n$. By Theorem 4.1 and Lemma 4.2, the normal cone has only one left coefficient line to recover: the entire line $\mathbb{C}\beta_R$. The canonical duality

$$\bigwedge^{N-2} W^* \simeq \bigwedge^2 W \otimes (\det W)^* = \bigwedge^2 W \otimes (\det V)^{-(n+1)} \quad (4.8)$$

sends that line to the Pluecker line of R , with the same projective transformation g . Indeed the pullback of a quotient volume is the annihilator-volume dual of the kernel R . The common determinant character is a scalar on projective space. A nonzero decomposable two-vector recovers its plane as $\{w \in W : w \wedge w_R = 0\}$. Thus an abstract scheme isomorphism implies $R' = gR$.

Conversely such a linear coordinate change induces the quotient isomorphism $S_R \rightarrow S_{R'}$, the algebra isomorphism $A_R \rightarrow A_{R'}$, and the corresponding Grassmannian isomorphism. It intertwines multiplication and hence preserves its zeroth Fitting ideal. This proves the reverse implication. The assertion about reductions was proved in (4.4). $\square$

4.3. Discriminants, elementary divisors, and hyperelliptic moduli. A regular pencil means one containing an invertible symmetric matrix; it need not have a reduced discriminant. The classical invariant of such a pencil consists of its discriminant points together with their Jordan-block partitions, usually called its Weierstrass–Segre data. We use the root-labelled partitions, not a pairing of an unlabelled list with an independently unlabelled divisor. Their classification is classical; compare [4, Theorem 1.1, Corollary 2.1 and §2]. The following consequence identifies precisely what the intrinsic failure scheme remembers beyond the discriminant.

Corollary 4.4 (Intrinsic recovery of elementary divisors). *For a regular pencil in any dimension $n \geq 3$, the abstract scheme $\widehat{D}_R$ determines all root-labelled elementary divisors of a matrix pencil, up to projective reparametrization of its parameter line. Conversely these data determine its abstract failure scheme. In particular the invariant distinguishes different Jordan partitions over a fixed multiple discriminant root.*

Proof. The first implication follows by reconstructing R , then choosing an invertible member A and a complementary member B . For completeness, the relevant classical sufficiency can be seen directly. Put $C = A^{-1}B$; it is self-adjoint for A . If $hC = C'h$ identifies the Jordan forms for another pair (A', B') , set

$$H = A^{-1}h^t A' h.$$

Then H is invertible, A -self-adjoint, and commutes with C . There is a polynomial k in H with $k^2 = H$: on each nonzero eigenvalue choose a square root and truncate its Taylor series to the size of the largest Jordan block, then combine by the Chinese remainder theorem. Thus k is invertible, self-adjoint, and commutes with C . The map hk^{-1} is an isometry from A to A' and intertwines C, C' , so it simultaneously identifies A, B with A', B' .

Changing the basis of the pencil replaces C by a fractional linear expression with invertible denominator. At each eigenvalue its derivative is nonzero; hence each Jordan-block size is preserved while its root moves by the same projective transformation. A parameter change can avoid infinity for both finite spectra. Root-labelled elementary divisors therefore suffice, and (4.7) proves the converse for the failure schemes. This argument recalls the classical classification; the new input is recovery from an unmarked abstract failure scheme. $\square$

Example 4.5 (Identical discriminant, distinct intrinsic failure schemes). Choose distinct nonzero $\mu_3, \dots, \mu_n$, and put

$$S_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad J_2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Consider the symmetric pairs

$$\begin{aligned} A_s &= I_n, & B_s &= \text{diag}(0, 0, \mu_3, \dots, \mu_n), \\ A_j &= S_2 \oplus I_{n-2}, & B_j &= S_2 J_2 \oplus \text{diag}(\mu_3, \dots, \mu_n). \end{aligned}$$

Their determinant binary forms agree up to a nonzero scalar:

$$\det(sA_s + tB_s) = -\det(sA_j + tB_j) = s^2 \prod_{i=3}^n (s + \mu_i t). \quad (4.9)$$

At the unique double root the ranks are respectively $n-2$ and $n-1$, so no simultaneous projective equivalence exists. The two root partitions are $(1, 1)$ and (2) . Their abstract failure schemes are therefore nonisomorphic although both their reduced failure schemes and their discriminant divisors, including multiplicities, agree. This realizes inside the same multiplication construction the classical loss of elementary-divisor information under passage to the determinant.

Corollary 4.6 (Hyperelliptic isomorphism classes). *Fix $g \geq 2$ and $n = 2g + 2$. On the locus of pencils with simple discriminant, $\widehat{D}_R$ gives an injective invariant of the isomorphism classes of smooth hyperelliptic curves of genus g . All these failure schemes have the same reduction; their nonreduced structures distinguish a $(2g - 1)$ -dimensional family of curve classes. The pencil invariant also distinguishes the two regular boundary pencils in Example 4.5, where their singular discriminant double covers alone do not.*

Proof. The classical curve associated with a simple-discriminant pencil is the double cover of its parameter line branched at $\det(sA + tB) = 0$; see [6, §4.2, equation (4.3)]. Choose an invertible member. Distinct eigenvalues of $A^{-1}B$ have mutually A -orthogonal eigenlines, each nonisotropic, so simultaneous diagonalization identifies the pencil with $\langle \sum x_i^2, \sum \lambda_i x_i^2 \rangle$. The unordered branch divisor modulo PGL_2 therefore determines the pencil, and every reduced divisor of degree $2g + 2$ arises. A smooth curve of genus at least two has a unique degree-two map to a rational curve up to automorphism of that curve: two distinct such maps would give a birational map to a curve of bidegree $(2, 2)$ in $\mathbb{P}^1 \times \mathbb{P}^1$, whose arithmetic genus is one. Thus curve isomorphism is exactly branch-divisor equivalence. Applying Theorem 4.3 gives the asserted injectivity and well-definedness.

Ordered distinct branch points have dimension $2g + 2$. Normalizing three of them removes the three-dimensional projective action; taking the finite permutation quotient leaves dimension $2g - 1$. Equality of all reductions follows from (4.4). Finally the two binary forms in (4.9) differ only by a scalar, a square over $\mathbb{C}$, and define isomorphic double covers, whereas their failure schemes have already been separated. $\square$

These are statements about complex isomorphism classes. They do not identify moduli stacks, assert a universal family over a coarse moduli space, or eliminate the different automorphism groups of pencils and curves. Nor is the classical pencil–hyperelliptic correspondence a new Torelli theorem. The conclusion here is reconstruction of those objects and of the elementary-divisor information on their boundary from the abstract nonreduced multiplication-failure scheme, while all reduced failure data are fixed.

5. THE EXACT INFINITESIMAL ORDER OF PENCIL RECONSTRUCTION

The all-pencil theorem admits a finite-order strengthening. It identifies the first infinitesimal neighbourhood that retains the pencil, and shows that every preceding neighbourhood is independent of it. We use ideal-adic neighbourhoods of the deepest stratum; these are not arc spaces or jet schemes.

Throughout this section let $n \geq 3$, $W = \mathrm{Sym}^2 V$, and

$$N = \binom{n+1}{2}, \quad p = N - 2, \quad d = n + 2p = n^2 + 2n - 4. \quad (5.1)$$

For a pencil $R \subset W$, put $S_R = W/R$ and $B = \mathrm{Gr}(n, S_R) \subset \widehat{D}_R$. This notation for the socle Grassmannian is local to this section; the cube-zero algebra is denoted by A_R , as in Section 4. Write $\mathfrak{b} \subset \mathcal{O}_{\widehat{D}_R}$ for the ideal of B , and define

$$Z_R^{[k]} = (B, \mathcal{O}_{\widehat{D}_R} / \mathfrak{b}^{k+1}) \quad (k \geq 0). \quad (5.2)$$

Thus $Z_R^{[0]} = B$. No embedding or identification of its reduction is supplied with the abstract scheme $Z_R^{[k]}$.

Theorem 5.1 (Sharp finite-neighbourhood reconstruction). *For any two pencils $R, R' \subset \text{Sym}^2 V$, the following conditions are equivalent:*

$$\begin{aligned} Z_R^{[d]} \simeq Z_{R'}^{[d]} &\iff \widehat{D}_R \simeq \widehat{D}_{R'} \\ &\iff R' = gR \text{ for some } g \in \text{PGL}(V). \end{aligned} \quad (5.3)$$

In contrast, for every $0 \leq k < d$, all the schemes $Z_R^{[k]}$ are isomorphic. Consequently d is the smallest uniform order at which these neighbourhoods classify all pencils up to projective equivalence. Singular pencils are included.

5.1. The first relation in the conormal algebra. Let $\mathcal{U}$ be tautological on B . The open set of X_R where projection to S_R is injective is canonically $\text{Tot } E$, where $E = \text{Hom}(\mathcal{U}, V)$: a plane is the graph of a unique map from its image in S_R to V . In particular this is one open neighbourhood of all of B , not merely a collection of unrelated formal charts. Let $\mathfrak{a}$ be its zero-section ideal. The exact multiplication calculation (4.4) gives

$$\mathcal{I}_R = (\det T) I_p(\gamma_R \text{Sym}^2 T) \subset \text{Sym } E^*. \quad (5.4)$$

It is generated in degree d , with a nonzero degree- d piece K_R . Indeed each quadratic minor has degree $2p$, and an invertible T makes $\gamma_R \text{Sym}^2 T$ surjective. There are no terms of lower degree, even on special pencils. Homogeneity is an equality of ideals in the actual graph neighbourhood, not just an equality of initial supports.

Lemma 5.2 (Intrinsic recovery from the first relation). *The abstract scheme $Z_R^{[d]}$ determines, as a graded $\mathcal{O}_B$ -algebra, the complete normal-cone algebra $\text{gr}_{\mathfrak{b}} \mathcal{O}_{\widehat{D}_R}$.*

Proof. The reduction of $Z_R^{[d]}$ is B , since its zero-section ideal is nilpotent and B is reduced. Hence its nilradical $\mathfrak{n}$ is intrinsic. Put $L = \mathfrak{n}/\mathfrak{n}^2$. Formula (5.4) and $d > 1$ identify $L = E^*$. Multiplication in the given finite scheme recovers

$$K_R = \ker[\text{Sym}^d L \longrightarrow \mathfrak{n}^d/\mathfrak{n}^{d+1}]. \quad (5.5)$$

The last denominator is zero, but retaining it displays the grading. No splitting of the reduction map is needed to form this kernel. Because $\mathcal{I}_R$ is generated in this single degree,

$$\text{gr}_{\mathfrak{b}} \mathcal{O}_{\widehat{D}_R} = \text{Sym}_{\mathcal{O}_B} L / (K_R). \quad (5.6)$$

Here (K_R) denotes the homogeneous ideal generated by K_R , not its radical. The right side is intrinsic to the finite scheme, proving the assertion. $\square$

Proof of Theorem 5.1. For $k < d$, equations (5.4) and (5.2) give

$$\mathcal{O}_{Z_R^{[k]}} = \text{Sym } E^* / \mathfrak{a}^{k+1}.$$

A linear identification $S_R \simeq S_{R'}$ identifies B , its tautological bundle, and E . It therefore identifies every one of these lower-order neighbourhoods.

An abstract isomorphism at order d induces an isomorphism of reductions, of conormal bundles, and, by Lemma 5.2, of complete graded normal cones. We spell out why the reconstruction in Theorem 4.1 requires no information outside this cone. Its reduced support is the square-matrix determinant cone; its iterated singular loci give the rank-one cone. The relative scheme of maximal linear spaces in that cone has the two rulings $\mathbb{P}(V) \times B$ and $\mathbb{P}_B(\mathcal{U}^*)$. Exactly the first is projectively trivial, as is seen by restriction to a Schubert line. Thus it is intrinsically oriented. The induced projective transformation of the trivial factor is constant on the connected projective base. Local lifts have the form $T \mapsto gTh^{-1}$ for this one $[g]$. The determinant ideal line is intrinsic, so its ideal quotient in (5.4) recovers the residual ideal. Finally Lemma 4.2 identifies the entire left coefficient line $\mathbb{C}\beta_R$, and exterior duality (4.8) recovers the Pluecker line of R . These are

operations on the recovered normal cone; we do not assume that its isomorphism extends to X_R or $\widehat{D}_R$. They imply $R' = gR$.

Conversely such a projective transformation induces the algebra and failure-scheme isomorphisms, carrying the socle Grassmannian to the socle Grassmannian and preserving all its ideal powers. This proves (5.3), also using Theorem 4.3. For sharpness, there are inequivalent pencils in every dimension under consideration: $\langle x_1^2, x_1x_2 \rangle$ has a common linear factor, whereas $\langle x_1^2, x_2^2 \rangle$ does not. Their neighbourhoods are nevertheless isomorphic at every order below d . Thus no smaller order classifies all pencils. $\square$

5.2. A closed immersion of the full parameter scheme. The finite-order result has a scheme-theoretic parameter version, which records infinitesimal variation as well as closed points. Here tensor coordinates are fixed. This additional qualification is important: we are not asserting an equivalence of unmarked moduli stacks.

Fix an auxiliary n -dimensional vector space U , and put

$$F(V) = \bigwedge^p \text{Sym}^2 V, \quad F(U) = \bigwedge^p \text{Sym}^2 U, \quad M = \dim F(U) = \binom{N}{2}, \quad H = \text{Hom}(U, V).$$

Choose a nonzero scalar representative δ of the determinant polynomial on H ; its scalar has no effect on the subspaces below. Let $K_R \subset \text{Sym}^d H^*$ be the degree- d space of $(\det T)I_p(\gamma_R \text{Sym}^2 T)$ in these fixed coordinates.

Theorem 5.3 (Faithful finite-order parameterization). *The assignment*

$$\Phi : \text{Gr}(2, \text{Sym}^2 V) \longrightarrow \text{Gr}(M, \text{Sym}^d H^*), \quad R \longmapsto K_R, \quad (5.7)$$

is a closed immersion. It is defined on the whole Grassmannian, including singular pencils. Its construction and its inverse on the image commute with arbitrary base change over $\mathbb{C}$. In particular it separates families over nonreduced bases and has injective differential at every pencil.

Lemma 5.4 (Tensoring a line with a fixed space). *For finite-dimensional vector spaces A, B_0 , with $b = \dim B_0 > 0$, the map $\mathbb{P}(A) \rightarrow \text{Gr}(b, A \otimes B_0)$, $\ell \mapsto \ell \otimes B_0$, is a closed immersion.*

Proof. Choose bases $a_0, \dots, a_t$ and a basis of B_0 . Above the projective chart $\ell = \mathbb{C}(a_0 + \sum x_i a_i)$, the subspace is the graph of the block matrices $x_i I_b$. The corresponding Grassmannian chart has arbitrary $b \times b$ blocks. The image is cut out by their off-diagonal entries and the differences between their diagonal entries; an inverse reads one diagonal entry in each block. These are scheme-theoretic equations over any $\mathbb{C}$ -algebra, not just pointwise conditions. The preimage of this Grassmannian chart is exactly the stated projective chart, since the projection has determinant a_0^b . Such charts cover the image. The map is proper because its source is projective, so these chartwise closed immersions give a global closed immersion. $\square$

Proof of Theorem 5.3. The coefficient map (4.6) becomes

$$\mu : F(V)^* \otimes F(U) \longrightarrow \text{Sym}^{2p} H^*.$$

It is injective: by Lemma 4.2 its source is irreducible for $\text{GL}(V) \times \text{GL}(U)$, and evaluation at an isomorphism $U \simeq V$ shows that the equivariant map is nonzero. Multiplication by the nonzero polynomial δ is injective. Thus $\nu = \delta\mu$ is a fixed linear inclusion and

$$K_R = \nu(\mathbb{C}\beta_R \otimes F(U)). \quad (5.8)$$

In particular $\dim K_R = M$, also on every singular pencil. By exterior duality the quotient-volume map $R \mapsto \mathbb{C}\beta_R$ is the Pluecker closed immersion $\text{Gr}(2, W) \hookrightarrow \mathbb{P}(F(V)^*)$. Factor Φ through this map, the closed immersion of Lemma 5.4, and the Grassmannian closed immersion induced by ν . This proves the theorem. All are morphisms of schemes defined by the displayed polynomial and

linear equations. A closed immersion is a monomorphism and remains one after arbitrary base change. This proves the family assertion, not merely separation of complex-valued points. $\square$

Corollary 5.5 (A flat family of finite normal algebras). *Over $\mathrm{Gr}(2, W)$, the algebras*

$$Q_R = \mathrm{Sym} H^* / ((K_R) + \mathfrak{m}^{d+1}), \quad \mathfrak{m} = \bigoplus_{j>0} \mathrm{Sym}^j H^*, \quad (5.9)$$

form a finite locally free family of rank

$$\binom{n^2 + d}{d} - \binom{N}{2}.$$

In fixed coordinates this family determines every base-changed family of pencils. All its truncations below degree d are constant families.

Proof. Theorem 5.3 makes the spaces K_R a rank- M subbundle of the constant degree- d bundle, with locally free quotient. In degree $j < d$ the algebra is $\mathrm{Sym}^j H^*$, and its degree- d piece is $\mathrm{Sym}^d H^* / K_R$; higher degrees vanish. This proves local freeness and base-change compatibility. Summing the ranks gives the displayed binomial identity. The kernel of the coordinate quotient in degree d recovers K_R , and Φ recovers the pencil family. This is a statement about coordinate quotient algebras; it does not discard their coordinate maps and then infer a family reconstruction theorem. $\square$

5.3. Classical moduli at a sharp order.

Corollary 5.6 (Hyperelliptic classes in finite neighbourhoods). *Let $g \geq 2$, $n = 2g + 2$, and $d = 4g^2 + 12g + 4$. For pencils with simple discriminant, $Z_R^{[d]}$ and $Z_{R'}^{[d]}$ are isomorphic if and only if their associated hyperelliptic curves are isomorphic. Every lower-order neighbourhood is independent of the curve. Thus the same $(2g - 1)$ -dimensional family of curve classes is invisible to all these lower-order neighbourhoods and is separated at order d .*

Proof. Combine Theorem 5.1 with Corollary 4.6, and substitute $n = 2g + 2$ in (5.1). $\square$

The order here concerns a neighbourhood of the entire projective socle Grassmannian. Its global ruling distinguishes the constant V -factor from the tautological factor. A single unmarked Artinian normal slice does not supply this orientation; the coordinate statement of Corollary 5.5 is deliberately different. The result is therefore not a claim that one unmarked local algebra replaces the projective family. The matrix-pencil classification and the hyperelliptic correspondence remain classical. The assertion proved here is that this classical information first appears at the specified order in the intrinsic multiplication-failure neighbourhoods.

6. RECONSTRUCTION ON A PROJECTIVE CURVE

The global part of the inverse argument has a more economical geometric realization. It uses a proper base to make the projective left frame constant, and a nontrivial right projective bundle to distinguish the rulings. Neither step requires the whole socle Grassmannian. We now replace that Grassmannian by a curve. In particular, a neighbourhood supported on one Schubert line already retains every pencil.

Let C be a smooth connected projective complex curve, and let $\mathcal{U}$ be a rank- n bundle on C , where $n \geq 3$. Assume that $\mathbb{P}_C(\mathcal{U}^*)$ is not projectively trivial. A sufficient numerical condition is

$$\deg \det \mathcal{U} \not\equiv 0 \pmod{n}. \quad (6.1)$$

Indeed a projectively trivial rank- n bundle is a line-bundle twist of a trivial bundle, so its determinant degree is divisible by n . Fix V of dimension n , put $E = \mathrm{Hom}(\mathcal{U}, V)$, and let $\mathfrak{a}$ be the

zero-section ideal in $\mathrm{Sym} E^*$. For a pencil $R \subset \mathrm{Sym}^2 V$, the quotient $\gamma_R : \mathrm{Sym}^2 V \rightarrow S_R$ defines the homogeneous ideal

$$\mathcal{I}_{R,C} = (\det T) I_p(\gamma_R \mathrm{Sym}^2 T), \quad p = \binom{n+1}{2} - 2, \quad (6.2)$$

where $T : \mathcal{U} \rightarrow V$ is the universal map on $\mathrm{Tot} E$. The expression is independent of local frames: both determinant multiplication and the maximal-minor construction are bundle constructions. Define

$$Y_{R,C}^{[k]} = \mathrm{Spec}_C(\mathrm{Sym} E^* / (\mathcal{I}_{R,C} + \mathfrak{a}^{k+1})).$$

Only its abstract complex scheme is retained, not its projection to C , its embedding in $\mathrm{Tot} E$, or either tensor factor.

Theorem 6.1 (Curve-supported reconstruction). *For fixed $(C, \mathcal{U}, V)$ as above and $d = n^2 + 2n - 4$, every pair of quadratic pencils satisfies*

$$Y_{R,C}^{[d]} \simeq Y_{R',C}^{[d]} \iff R' = gR \text{ for some } g \in \mathrm{PGL}(V). \quad (6.3)$$

All lower-order neighbourhoods are independent of R . The order- d schemes are finite locally free over C of rank $\binom{n^2+d}{d} - \binom{N}{2}$, where $N = \binom{n+1}{2}$. No regularity condition is imposed on the pencil.

Proof. On any frame chart, the ideal in (6.2) is generated in degree d . Its degree- d piece K_R is a subbundle of rank $\binom{N}{2}$, by the fixed coefficient inclusion (5.8); its quotient is locally free. The quotient algebra has its full symmetric powers in degrees less than d , has $\mathrm{Sym}^d E^* / K_R$ in degree d , and is zero above that degree. This proves the rank and the assertion about lower orders.

Its reduction is C , and its nilradical $\mathfrak{n}$ recovers $L = \mathfrak{n}/\mathfrak{n}^2$ and

$$K_R = \ker[\mathrm{Sym}^d L \rightarrow \mathfrak{n}^d].$$

Thus the abstract finite-order scheme reconstructs the entire graded algebra $\mathrm{Sym} L / (K_R)$, by the same argument as Lemma 5.2. The ideal here means the homogeneous ideal generated by the recovered kernel, not its radical. No splitting of the reduction or supplied projection is used.

The reduction of this reconstructed cone is the determinant cone. Indeed its ideal has the determinant factor, and when T is invertible the quadratic quotient has full rank. Its rank-one Fano scheme has two ruling components, $\mathbb{P}(V) \times C$ and $\mathbb{P}_C(\mathcal{U}^*)$. The hypothesis distinguishes the first one intrinsically. An abstract isomorphism of neighbourhoods may induce a nonidentity automorphism of C ; pull back the second bundle along that automorphism. The oriented ruling map is nevertheless a morphism from C to PGL_n . It is constant, since C is connected projective and $H^0(C, \mathcal{O}_C) = \mathbb{C}$. Lemma 2.2 therefore supplies one constant projective left frame.

Cancel the intrinsic determinant ideal and apply the coefficient construction of Theorem 3.1. For pencils, Lemma 4.2 leaves just the single line $\mathbb{C}\beta_R$. Exterior duality (4.8) recovers R with respect to that same projective frame. This proves the forward implication in (6.3). Conversely a linear lift of g induces the bundle map on E and the quotient isomorphism on S_R , and carries all the displayed ideals and their truncations to the corresponding ideals for R' . This proves the converse. $\square$

Corollary 6.2 (One Schubert line suffices). *Let $\ell \simeq \mathbb{P}^1$ be any Schubert line in $B_R = \mathrm{Gr}(n, S_R)$, and use the canonical graph projection $Z_R^{[d]} \rightarrow B_R$ to form its restriction $Z_{R,\ell}^{[d]}$. Its abstract isomorphism class is independent of the choice of ℓ , and*

$$Z_{R,\ell}^{[d]} \simeq Z_{R',\ell'}^{[d]} \iff R' = gR \text{ for some } g \in \mathrm{PGL}(V).$$

The smallest uniform reconstruction order for these curve-supported neighbourhoods is again d . Their reductions all have dimension one, instead of $n(p-n)$.

Proof. The tautological bundle on a Schubert line is $\mathcal{O}_{\mathbb{P}^1}(-1) \oplus \mathcal{O}_{\mathbb{P}^1}^{n-1}$. Pulling back the homogeneous ideal of the graph neighbourhood gives exactly (6.2) with that bundle. Its isomorphism type depends on this bundle and R , not on the chosen inclusion of the line in the socle. The determinant degree is -1 , so Theorem 6.1 applies. Its lower-order schemes are all independent of the pencil, while $\langle x_1^2, x_1x_2 \rangle$ and $\langle x_1^2, x_2^2 \rangle$ are inequivalent. This proves the uniform minimality assertion. $\square$

This restriction is made in the construction; it is not a claim that an abstract neighbourhood canonically chooses a Schubert line. No line is supplied to an isomorphism in the corollary: the reduced curve is recovered as the reduction of its source. Nor is the corollary a statement about one unmarked Artinian normal slice. The projective curve and the nontrivial ruling remain essential to this argument. The same theorem applies, for example, to $\mathcal{U} = \mathcal{O}_C(-P) \oplus \mathcal{O}_C^{n-1}$ on every smooth projective connected curve with a point P . It isolates the properness and ruling obstruction used in reconstruction, independently of the original Grassmannian realization.

7. SPECTRAL TORSION AND RANK-PRESERVING SPECIALIZATION

The finite-neighbourhood theorem recovers more than the discriminant or the corank of each member of a regular pencil. We identify its classical spectral invariant and then give a flat specialization on which the entire discriminant and every reduced rank locus remain constant, while the recovered invariant changes. The two specializations are separated at exactly the order of Theorem 5.1. The parameter space throughout is the usual Grassmannian of pencils; the failure map is the original multiplication map, not an added coefficient-evaluation block.

7.1. The spectral torsion sheaf. Let $R \subset \text{Sym}^2 V$ be a regular pencil, meaning that it contains a nonsingular quadratic form. On its parameter line $L_R = \mathbb{P}(R)$, the tautological quadratic form gives an exact sequence

$$0 \longrightarrow V^* \otimes \mathcal{O}_{L_R}(-1) \xrightarrow{q_R} V \otimes \mathcal{O}_{L_R} \longrightarrow \mathcal{T}_R \longrightarrow 0. \quad (7.1)$$

The first arrow is injective because its determinant is nonzero at the generic point of the smooth integral curve L_R . Thus $\mathcal{T}_R$ is a torsion sheaf of length n , whose zeroth Fitting divisor is the full discriminant. The pair $(L_R, \mathcal{T}_R)$ is considered up to isomorphism of the parameter line. Its definition involves no choice of a basis of R . The relation with elementary divisors is classical; see [4, §2, Corollary 2.1 and equation (5)]. We record the argument to make precise which datum is recovered.

Lemma 7.1 (Sheaves on the affine spectral line). *Let A, B be symmetric complex matrices with A invertible. On the affine line with coordinate z , the cokernel of $zA - B$ is the finite-dimensional $\mathbb{C}[z]$ -module on $\mathbb{C}^n$ for which multiplication by z is $C = A^{-1}B$. An isomorphism of such sheaves, in this fixed coordinate, is equivalent to similarity of the corresponding matrices.*

Proof. Multiplication on the target by A^{-1} identifies the presentation with $zI - C$. Reduction of a polynomial vector modulo these relations sends $\sum_j z^j v_j$ to $\sum_j C^j v_j$; constant vectors give the inverse identification. The equivalence between coherent sheaves on an affine scheme and finitely generated modules identifies a finite-length sheaf supported in this affine line with this module. The support is finite, so its underlying complex vector space is finite dimensional. A module isomorphism is a linear isomorphism G satisfying $GC = C'G$, which is exactly similarity. Extension by zero across a point outside the support loses no sheaf data. $\square$

Lemma 7.2 (Polynomial square roots of invertible operators). *For every invertible complex matrix K there is $h \in \mathbb{C}[x]$ with $h(K)^2 = K$. If K is self-adjoint for a nondegenerate symmetric form and commutes with C , then $h(K)$ has both properties.*

Proof. Write the minimal polynomial as $\prod_{\lambda}(x - \lambda)^{a_{\lambda}}$, where $\lambda \neq 0$. In its local Artinian factor choose a square root $\mu_{\lambda}^2 = \lambda$, and put

$$h_{\lambda}(x) = \mu_{\lambda} \sum_{j=0}^{a_{\lambda}-1} \binom{1/2}{j} \left(\frac{x - \lambda}{\lambda} \right)^j.$$

The formal binomial identity, truncated modulo $(x - \lambda)^{a_{\lambda}}$, gives $h_{\lambda}(x)^2 = x$. The Chinese remainder theorem joins these polynomials to a single class modulo the minimal polynomial. Evaluation at K proves the assertion even when K is not semisimple. Polynomials in one self-adjoint operator remain self-adjoint, and every polynomial in K commutes with every operator commuting with K . $\square$

Theorem 7.3 (Spectral reconstruction at the first relation). *Let $n \geq 3$, $d = n^2 + 2n - 4$, and let R, R' be regular pencils. The following are equivalent:*

- (1) *the unmarked schemes $Z_R^{[d]}$ and $Z_{R'}^{[d]}$ are isomorphic;*
- (2) *there is an isomorphism $f : L_R \rightarrow L_{R'}$ with $\mathcal{T}_R \simeq f^* \mathcal{T}_{R'}$;*
- (3) *the pencils R, R' are projectively equivalent.*

In particular the finite scheme determines the full root-labelled spectral torsion sheaf, not only its Fitting divisor or its support. This is an equivalence of isomorphism classes of objects; it is not an assertion identifying their automorphism groups or moduli stacks.

Proof. The equivalence of the first and third conditions is Theorem 5.1. A congruence of pencils induces an isomorphism of (7.1), so the third implies the second.

For the converse choose a common affine coordinate whose point at infinity avoids the two spectral supports, using f to identify the lines. Write the corresponding symmetric pairs as (A, B) and (A', B') , with A, A' invertible, and set $C = A^{-1}B$, $C' = A'^{-1}B'$. Choose the presentation $zA - B$ for the affine coordinate on both lines. Lemma 7.1 identifies the two sheaves with the modules of C, C' , so their isomorphism gives a similarity. After applying that change of coordinates, assume $C' = C$. Both A and A' are nondegenerate symmetric forms for which C is self-adjoint. Then $K = A^{-1}A'$ commutes with C and is self-adjoint for A . Lemma 7.2 provides a polynomial h with $h(K)^2 = K$. Put $H = h(K)$. It commutes with C , is self-adjoint for A , and satisfies

$$H^t A H = A', \quad H^t B H = H^t A C H = A' C = B'.$$

Thus the pairs, and hence the pencils, are congruent. This also shows why the spectral sheaf alone suffices here; no additional pairing on that sheaf is required over $\mathbb{C}$. $\square$

For $x \in \text{Supp } \mathcal{T}_R$, put

$$\ell_a(x) = \text{length}(\mathcal{T}_{R,x} / \mathfrak{m}_x^a \mathcal{T}_{R,x}), \quad a \geq 0. \quad (7.2)$$

If the elementary-divisor exponents at x are $e_1, \dots, e_b$, the structure theorem over the discrete valuation ring $\mathcal{O}_{L_R,x}$ gives

$$\ell_a(x) = \sum_{j=1}^b \min(a, e_j), \quad \ell_a(x) - \ell_{a-1}(x) = \#\{j : e_j \geq a\}. \quad (7.3)$$

Thus these intrinsic local lengths recover every exponent, including repeated roots. The corank of the pencil at x gives only $\ell_1(x)$, and the multiplicity of the discriminant gives only the eventual value $\ell_a(x) = \sum_j e_j$. The intermediate values are genuinely additional information.

7.2. A specialization with constant discriminant and coranks. The following family keeps both ends of this length profile fixed. Fix $i \in \mathbb{C}$ with $i^2 = -1$, and set

$$N_0 = \begin{pmatrix} 1 & i \\ i & -1 \end{pmatrix}, \quad P_t = \frac{t}{2}N_0, \quad A_0 = \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix}, \quad C_t = \begin{pmatrix} P_t & I_2 \\ 0 & P_t \end{pmatrix}. \quad (7.4)$$

Put

$$B_t = A_0 C_t = \begin{pmatrix} 0 & P_t \\ P_t & I_2 \end{pmatrix}, \quad R_t = \langle A_0, B_t \rangle \subset \text{Sym}^2 \mathbb{C}^4. \quad (7.5)$$

Both matrices are symmetric and A_0 is invertible. They are independent for every t . For $n > 4$, append I_{n-4} to A_0 and append $\text{diag}(\mu_5, \dots, \mu_n)$ to B_t , where the μ_j are fixed distinct nonzero numbers. Denote the resulting matrices by $A^{(n)}, B_t^{(n)}$ and their pencil again by R_t .

Theorem 7.4 (Separation inside fixed reduced rank data). *For every $n \geq 4$, the family R_t , $t \in \mathbb{A}^1$, has the following properties.*

- (1) *Its full discriminant, with multiplicities, is independent of t :*

$$\det(sA^{(n)} + uB_t^{(n)}) = s^4 \prod_{j=5}^n (s + \mu_j u). \quad (7.6)$$

For every minor size, the reduced determinantal locus on the marked parameter line is also independent of t .

- (2) *At the root $s = 0$, the spectral module has exponents $(3, 1)$ for $t \neq 0$ and $(2, 2)$ for $t = 0$. Consequently $Z_{R_t}^{[d]} \not\cong Z_{R_0}^{[d]}$ for every $t \neq 0$, whereas all the punctured fibres are isomorphic.*
- (3) *The schemes $Z_{R_t}^{[d]}$ form a flat projective family over $\mathbb{A}^1$, with fixed reduction. After one identification of the socle quotient bundles, every truncation of order $k < d$ is a constant family. Thus the first separating order remains exactly $d = n^2 + 2n - 4$ even under the constraints in part (1).*

Proof. The identities $N_0^2 = 0$ and $\text{rank } N_0 = 1$ give

$$C_t^2 = \begin{pmatrix} 0 & tN_0 \\ 0 & 0 \end{pmatrix}, \quad C_t^3 = 0, \quad C_t = \begin{pmatrix} I_2 \\ P_t \end{pmatrix} \begin{pmatrix} P_t & I_2 \end{pmatrix}. \quad (7.7)$$

The right factor in the last expression is surjective and the left factor is injective. Hence $\text{rank } C_t = 2$ for every t , whereas $\text{rank } C_t^2 = 1$ precisely when $t \neq 0$. The nilpotent partitions are therefore $(3, 1)$ and $(2, 2)$. Also $\det A_0 = 1$ and $\det(sI_4 + uC_t) = s^4$, proving (7.6). At $s = 0$ the full pencil has corank two for all t ; at each remaining root it has corank one, and elsewhere corank zero. Equality of these functions on $\mathbb{P}^1(\mathbb{C})$ implies equality of the reduced closed determinantal subschemes for each minor size.

At the unique multiplicity-four root, formula (7.3) yields respectively

$$(\ell_1, \ell_2, \ell_3, \dots) = (2, 3, 4, \dots), \quad (2, 4, 4, \dots).$$

No automorphism of the parameter line can exchange this root with a simple one. Theorem 7.3 separates the special fibre from the punctured fibres and identifies the latter with one another. This is a statement about geometric fibres; no choice of a simultaneous trivializing congruence on $\mathbb{A}^1 \setminus 0$ is needed.

For flatness we use the relative gluing of Theorem 8.1; the following coordinates identify its pullback in the present family. First note that $t \mapsto R_t$ is a morphism to $\text{Gr}(2, \text{Sym}^2 V)$, not merely a collection of pencils. In symmetric matrix coordinates the two fixed linear functionals

$$a(Q) = Q_{13} + iQ_{14}, \quad b(Q) = Q_{33}$$

satisfy $a(A^{(n)}) = b(B_t^{(n)}) = 1$ and $a(B_t^{(n)}) = b(A^{(n)}) = 0$. Their common kernel is a fixed complement to every R_t . It trivializes $S_{R_t} = \text{Sym}^2 V/R_t$, and hence identifies the relative socle Grassmannian with $B \times \mathbb{A}^1$.

On a frame chart of its tautological bundle, the graph description (5.4) identifies the truncated algebra with

$$\bigoplus_{j=0}^{d-1} \text{Sym}^j E^* \oplus (\text{Sym}^d E^*/K_{R_t}). \quad (7.8)$$

By Theorem 5.3, K_{R_t} is a subbundle of constant rank $\binom{N}{2}$, with locally free quotient. This statement commutes with pullback to the present base and with right changes of frame, so the local descriptions glue. Thus (7.8) is locally free on $B \times \mathbb{A}^1$, of rank $\binom{n^2+d}{d} - \binom{N}{2}$. The graph projection is a finite flat morphism to that base. Since B is projective, its composition with projection to $\mathbb{A}^1$ is flat and projective. Below degree d there is no relation, and the fixed complement identifies the entire lower truncated algebra, including multiplication, with a constant family. These arguments prove flatness of the finite neighbourhoods; they make no assertion about flatness of the unrestricted failure schemes. $\square$

The qualifier “reduced” in part (1) is necessary. At $s = 0$, in a local parameter $z = s/u$, the gcd of the three-row minors of the four-dimensional block is a unit times z for $t \neq 0$ and a unit times z^2 for $t = 0$. Both radicals are (z) . Thus the family retains exactly the higher Fitting information discarded by reduction; it does not contradict the classical classification by the full determinantal ideals.

7.3. An independently visible change of reciprocal geometry. For a regular pencil let its reciprocal curve be the projective closure of the inverses of its nonsingular members. These curves and their strata are studied independently of multiplication-failure schemes in [4, §3].

Corollary 7.5 (A conic specializes to a line). *For the four-dimensional family (7.5), the reciprocal curve is a smooth plane conic when $t \neq 0$ and a projective line when $t = 0$. The unmarked twentieth neighbourhood distinguishes these two cases, while all earlier neighbourhoods, the full discriminant, and all reduced rank loci agree.*

Proof. For $s \neq 0$, multiplication on the right by the fixed invertible matrix A_0^{-1} identifies the projective inverse with

$$(sI_4 + uC_t)^{-1} = s^{-1}I_4 - us^{-2}C_t + u^2s^{-3}C_t^2.$$

If $t \neq 0$, the matrices I_4, C_t, C_t^2 are linearly independent: their span has dimension equal to the degree three of the minimal polynomial. The closure is therefore the quadratic Veronese image $[s : u] \mapsto [s^2 : -su : u^2]$, a smooth conic in its spanning plane. At $t = 0$ the last term vanishes, and I_4, C_0 are independent, giving a line. The remaining assertions follow from Theorem 7.4, with $d = 20$. The degree drop does not assert that the reciprocal curves themselves form a flat family; the flat family proved above consists of the finite failure neighbourhoods. $\square$

8. THE UNIVERSAL FINITE NEIGHBOURHOOD AND SPECTRAL STRATA

The finite-order construction is a family of actual neighbourhoods in multiplication-failure schemes. This fact is needed when passing from individual pencils to degenerations and infinitesimal parameter spaces. We establish it before discussing the spectral incidence schemes.

Put $P = \text{Gr}(2, \text{Sym}^2 V)$, and write $\mathcal{R}$ and $\mathcal{S} = (\text{Sym}^2 V \otimes \mathcal{O}_P)/\mathcal{R}$ for its universal relation and quotient bundles. Let

$$\pi : \mathcal{B} = \text{Gr}_P(n, \mathcal{S}) \longrightarrow P, \quad \mathcal{E} = \text{Hom}(\mathcal{U}, V \otimes \mathcal{O}_{\mathcal{B}}),$$

where $\mathcal{U}$ is tautological of rank n . As before, $N = n(n+1)/2$, $p = N - 2$, $d = n + 2p$, and $M = \binom{N}{2}$.

Theorem 8.1 (Universal neighbourhood and arbitrary base change). *The universal multiplication-failure scheme has a canonical order- d neighbourhood $\mathcal{Z}^{[d]}$ of $\mathcal{B}$. There is a rank- M vector subbundle $\mathcal{K} \subset \mathrm{Sym}^d \mathcal{E}^*$ such that*

$$\mathcal{Z}^{[d]} = \mathrm{Spec}_{\mathcal{B}} \left(\mathrm{Sym} \mathcal{E}^* / (\mathcal{K}, \mathrm{Sym}^{d+1} \mathcal{E}^*) \right). \quad (8.1)$$

This is finite locally free over $\mathcal{B}$, of rank

$$\binom{n^2 + d}{d} - \binom{N}{2}, \quad (8.2)$$

and is projective and flat over P . Its construction, its ideal-adic filtration, and its identification with the actual multiplication-failure neighbourhood commute with every morphism of complex schemes $T \rightarrow P$, including nonreduced ones. For $k < d$, its order- k truncation is $\mathrm{Spec}_{\mathcal{B}} \bigoplus_{j=0}^k \mathrm{Sym}^j \mathcal{E}^$, with truncated multiplication. After trivializing $\mathcal{S}$ on the parameter base, these lower-order families are constant.*

Proof. The locus in $\mathrm{Gr}_P(n, V \oplus \mathcal{S})$ on which projection to $\mathcal{S}$ is injective is canonically $\mathrm{Tot} \mathcal{E}$. It is an open neighbourhood of $\mathcal{B}$, not a choice of formal coordinates. On a frame chart of $\mathcal{U}$, let $T : \mathcal{U} \rightarrow V$ be the universal graph map and let $\gamma : \mathrm{Sym}^2 V \rightarrow \mathcal{S}$ be the quotient. The same block elimination that proves (5.4) works over its coordinate ring: the ideal of the multiplication-failure scheme is

$$(\det T) I_p(\gamma \mathrm{Sym}^2 T).$$

Every generator is homogeneous of degree d .

Here is also a coordinate-free description of its first relation bundle. The coefficient map, followed by determinant multiplication, is a morphism

$$\det V^* \otimes \det \mathcal{U} \otimes \det \mathcal{S}^* \otimes \bigwedge^p \mathrm{Sym}^2 \mathcal{U} \longrightarrow \mathrm{Sym}^d \mathcal{E}^*. \quad (8.3)$$

The inclusion used in this expression is $\det \mathcal{S}^* \hookrightarrow (\bigwedge^p \mathrm{Sym}^2 V)^*$. The determinant factors evaluate $\det T$, and the remaining factors evaluate $\bigwedge^p \mathrm{Sym}^2 T$ followed by the quotient. Thus the image is exactly the span of the displayed maximal minors, not a separately inserted coefficient module.

On a frame chart this map is the fixed injective linear coefficient map ν of Theorem 5.3, restricted to a line subbundle tensored with the fixed space $\bigwedge^p \mathrm{Sym}^2 \mathbb{C}^n$. Its image is a subbundle of rank M : locally split the line in the source, and split the fixed complex linear inclusion ν . Both splittings remain splittings over any complex algebra. Denote its image by $\mathcal{K}$. A right change of frame of $\mathcal{U}$ changes the graph coordinates and this coefficient map contragrediently. The determinant character in (8.3) is exactly the one arising from $\det T$. Hence these local ideals and their degree- d subbundles agree on overlaps. In particular, they glue as ideals of the given graph neighbourhood.

Let $\mathfrak{b}$ be the ideal of the zero section inside the multiplication-failure scheme. In graph coordinates its preimage is the positive-degree ideal. Quotienting by $\mathfrak{b}^{d+1}$ therefore gives precisely (8.1). This identifies the gluing with the actual ideal-adic neighbourhood, not merely with algebras having the correct geometric fibres. Its underlying module is

$$\bigoplus_{j=0}^{d-1} \mathrm{Sym}^j \mathcal{E}^* \oplus (\mathrm{Sym}^d \mathcal{E}^* / \mathcal{K}).$$

All terms are locally free. Counting monomials in n^2 variables gives (8.2). The morphism $\mathcal{B} \rightarrow P$ is smooth and projective, so the asserted flatness and projectivity follow. Finally the graph

description, the split coefficient inclusion, and the homogeneous quotient all commute with base change. Below degree d the ideal contributes no relation. This proves the final assertion as well. $\square$

Remark 8.2. Over a nonreduced base, the theorem concerns the specified relative socle subscheme $\mathcal{B}_T$ and its ideal. It does not identify that ideal with the absolute nilradical of $\mathcal{Z}_T^{[d]}$, which can also contain nilpotents from the base. On geometric fibres $\mathcal{B}_t$ is exactly the reduction. Nor is this theorem an equivalence of unmarked deformation groupoids: extra automorphisms of truncated algebras are not being identified with pencil congruences.

8.1. Spectral incidence schemes from the first relation space. On the regular-pencil locus, pass locally to ordered generators (A, B) with A invertible, and put $C = A^{-1}B$. For a marked point with affine coordinate z , set $X = C - zI$. The next elementary presentation works over an arbitrary coefficient ring; no diagonalization or division by eigenvalue differences is involved.

Lemma 8.3 (Truncated spectral presentation). *Let D be a commutative ring, $X \in \text{Mat}_n(D)$, and $a \geq 1$. There is a natural isomorphism*

$$\text{coker}(wI - X) \otimes_{D[w]} D[w]/(w^a) \simeq \text{coker}(X^a : D^n \rightarrow D^n). \quad (8.4)$$

Equivalently, the a -square block presentation with diagonal blocks X and subdiagonal blocks $-I$ is equivalent, by invertible row and column operations over D , to $I_{n(a-1)} \oplus X^a$.

Proof. Map the class of $\sum_{j=0}^{a-1} w^j v_j$ to the class of $\sum_{j=0}^{a-1} X^j v_j$ modulo $X^a D^n$. The relation $wv = Xv$ and the relation $w^a v = 0$ show that this map is well-defined. Sending a constant vector to its class is inverse. Eliminating successively the blocks with coefficient $-I$ gives the stated presentation equivalence, using only unit pivots. This also proves naturality under ring homomorphisms. $\square$

Theorem 8.4 (Scheme-theoretic spectral readout). *In the fixed tensor presentation of Theorem 5.3, the first relation subspace determines, on the regular-pencil locus, every closed spectral incidence scheme*

$$\mathfrak{S}_{a,h} = V(I_{n-h+1}((C - zI)^a)), \quad 1 \leq a, h \leq n. \quad (8.5)$$

This determination commutes with arbitrary complex base change. At a geometric point, membership is equivalent to $\text{length}(\mathcal{T}_z/\mathfrak{m}_z^a \mathcal{T}_z) \geq h$. Consequently the schemes of specified spectral length profiles, and not only their reduced supports, can be read from the order- d relation subspace.

Proof. The closed immersion Φ of Theorem 5.3 has an inverse on its scheme-theoretic image. Thus a family of relation subspaces in that image recovers its morphism to P , over any base, and hence its universal pencil. In the displayed chart its spectral sheaf has presentation $(z+w)I - C = wI - X$. Lemma 8.3 gives the presentation X^a for restriction to the order- a fat point centred at z . Its $(h-1)$ -st Fitting ideal is exactly the ideal in (8.5); Fitting ideals of finitely presented modules commute with base change. On a field the cokernel has dimension at least h exactly when these minors vanish.

These descriptions are independent of the chosen frames. A change of basis in V conjugates X ; a change of affine coordinate replaces the parameter ideal at the marked point by a unit multiple locally. Alternatively, restriction of the spectral sheaf to the fat diagonal on its parameter line defines the same finite module without a coordinate. Thus the Fitting ideals glue. Rank equality is imposed by the corresponding rank inequality together with the open complement of the next inequality. Intersecting these locally closed conditions gives each specified length profile. Formula (7.3) then identifies its geometric fibres with the prescribed elementary-divisor partition. $\square$

8.2. Realizing the complete Segre stratification. Let P_σ denote the reduced closure in P of a regular Segre stratum, and let $J_\sigma = \Phi(P_\sigma)$. Because Φ is a closed immersion, it gives isomorphisms

$$P_\sigma \simeq J_\sigma, \quad P_{\sigma_1} \cap_P \cdots \cap_P P_{\sigma_s} \simeq J_{\sigma_1} \cap_{\Phi(P)} \cdots \cap_{\Phi(P)} J_{\sigma_s}, \quad (8.6)$$

where the intersections carry their scheme structures. In particular their local rings and formal completions are preserved. Pulling back Theorem 8.1 realizes these parameter schemes by flat projective families of the actual finite failure neighbourhoods. The classical inclusions among Segre-stratum closures [4, Theorem 5.1] are therefore all realized, rather than just the one degeneration of Theorem 7.4.

There are two distinct assertions here. The Segre closure order is classical, and transport of closed subschemes through a closed immersion is formal. The geometric input proved here is the base-change-compatible realization by actual multiplication-failure neighbourhoods and the recovery of their spectral Fitting schemes. The ambient parameter space in (8.6) is the image of Φ , not a moduli space of every possible unmarked deformation of a finite scheme.

9. FIXED-DISCRIMINANT STRATA AND ALL ADMISSIBLE SPECIALIZATIONS

The universal family of Section 8 and the example of Section 7 lead to a classification of the entire fixed-discriminant, fixed-reduced-rank locus. The spectral orbit order is classical. Every allowed specialization is realized here by a flat family of the original finite multiplication-failure neighbourhoods, and the same first relation detects every strict transition.

Fix distinct $\alpha_1, \dots, \alpha_s \in \mathbb{C}$, positive m_i with $\sum m_i = n \geq 3$, and $1 \leq c_i \leq m_i$. Fix a nonsingular symmetric A , and let $\mathfrak{p}_A$ be its self-adjoint endomorphisms. Consider the reduced locally closed variety

$$X_{m,c}(A) = \left\{ C \in \mathfrak{p}_A : \det(zI - C) = \prod_i (z - \alpha_i)^{m_i}, \quad \dim \ker(C - \alpha_i I) = c_i \right\}. \quad (9.1)$$

Exclude $s = 1, c_1 = n$, for which C is scalar and $\langle A, AC \rangle$ is not a pencil. Every other datum is realized. Put $R_C = \langle A, AC \rangle$, $d = n^2 + 2n - 4$, and $Z_C = Z_{R_C}^{[d]}$. The full binary determinant is $\det(A) \prod_i (u + \alpha_i v)^{m_i}$. The coranks and labelled roots specify all reduced determinantal loci on the line.

Write $\mathcal{P}(m, c)$ for partitions into exactly c positive parts. For equal-size partitions, $\mu \preceq \lambda$ means $\sum_{j \leq a} \mu_j \leq \sum_{j \leq a} \lambda_j$ for every a ; write λ' for the conjugate partition.

9.1. The classical orthogonal input. We isolate the orbit theorem used below, with its precise group and source. Let (H, A) be a nondegenerate complex symmetric space of dimension m . The acting group is the full orthogonal group $O(A)$, and the operators are A -self-adjoint, not elements of the skew-adjoint Lie algebra $\mathfrak{o}(A)$.

Lemma 9.1 (Self-adjoint nilpotent orbits). *The full $O(A)$ -orbits of self-adjoint nilpotent operators are indexed by all partitions λ of m . If $\mathcal{O}_\lambda$ denotes the orbit, then*

$$\dim \mathcal{O}_\lambda = \frac{1}{2} \left(m^2 - \sum_a (\lambda'_a)^2 \right), \quad \mathcal{O}_\mu \subset \overline{\mathcal{O}_\lambda} \iff \mu \preceq \lambda.$$

An $O(A)$ -orbit is one $SO(A)$ -orbit when λ has an odd part. When every part is even, it is the disjoint union of two $SO(A)$ -orbits, interchanged by an isometry of determinant -1 . The closure statement above is for the full orthogonal orbit and does not identify the closures of its two components.

Proof. The classification and closure assertions are [5, Proposition 1, p. 444; Theorem 1, p. 447], respectively, with $\varepsilon = 1$ and $G(V)$ as defined in [5, §1.1, p. 443]. Remark 1 on p. 444 specifies that all partitions occur; the order in §1.4 on p. 446 is precisely the row-dominance order used here. The proof of Theorem 1 occupies pp. 447–451. These references concern the full group, not its identity component.

For the dimension formula there is a short independent calculation. The trace pairing on $\text{End } H$ is nondegenerate on both its self-adjoint and skew-adjoint summands. For a self-adjoint nilpotent N , the commutator with N exchanges the summands, and the two resulting maps are adjoint up to sign for that pairing. They have equal rank. Consequently $\dim \text{GL}(H)N = 2 \dim O(A)N$, also the identity in [5, §2.4, Remark 8(i), p. 456]. The general linear centralizer has dimension $\sum_{i,j} \min(\lambda_i, \lambda_j) = \sum_a (\lambda'_a)^2$, as follows by writing homomorphisms between the cyclic $\mathbb{C}[t]$ -modules of the Jordan decomposition. This proves the stated dimension.

We give the component argument explicitly. The good bases of [5, Lemma 1, p. 444] decompose the symmetric space orthogonally into cyclic Jordan blocks, each with a reversed-identity Gram matrix. If one block has odd size, its negative identity, extended by the identity on the other blocks, centralizes N and has determinant -1 . The centralizer then meets both components of $O(A)$, so the orbit is a single $SO(A)$ -orbit.

For the converse, write

$$E_k = \ker N^k / (\ker N^{k-1} + N \ker N^{k+1}).$$

The form induced by $A(u, N^{k-1}v)$ on E_k is nondegenerate and symmetric; this is immediate on the same cyclic blocks. A commuting isometry induces $g_k \in O(E_k)$. A filtration of the Jordan module by its cyclic lengths and powers of t gives

$$\det g = \prod_k (\det g_k)^k.$$

Indeed the multiplicity-space action of a length- k block occurs k times on the associated graded; the strictly triangular terms of a module automorphism do not change its determinant. If all lengths are even, this product is one. The centralizer lies in $SO(A)$, and the index-two quotient of $O(A)$ therefore gives exactly two $SO(A)$ -orbits. Since $SO(A)$ is connected, these are the connected components of the full orbit. This does not preclude their closures from meeting. $\square$

Theorem 9.2 (Classification and specialization at the first relation). *The following statements hold.*

- (1) *The $O(A)$ -orbits in $X_{\mathbf{m}, \mathbf{c}}(A)$ are indexed by $\boldsymbol{\lambda} \in \prod_i \mathcal{P}(m_i, c_i)$, with dimensions*

$$\dim \mathcal{O}_{\boldsymbol{\lambda}} = \frac{1}{2} \left(n^2 - \sum_i \sum_a (\lambda'_{i,a})^2 \right). \quad (9.2)$$

Unmarked isomorphism classes of Z_C are these tuples modulo the permutation group induced on the roots by projective transformations preserving their multiplicities and coranks.

- (2) *A specialization from $\boldsymbol{\lambda}$ to $\boldsymbol{\mu}$ inside (9.1) exists if and only if*

$$\mu_i \preceq \lambda_i \quad \text{for every } i. \quad (9.3)$$

Every such pair is realized over a smooth pointed algebraic curve, after shrinking at the special point, with A fixed and with constant full determinant and reduced rank loci.

- (3) *Every such pencil family induces a flat projective family of Z_C . After shrinking the curve, every truncation of order $k < d$ is constant. If at least one dominance inequality is strict, the generic and special Z_C are nonisomorphic; their first distinguishing order is d .*

(4) *The recovered spectral module at α_i has length profile*

$$L_{i,a} = \sum_j \min(a, \lambda_{i,j}). \quad (9.4)$$

Every $L_{i,a}$ can only increase in the stated specialization. All these inequalities together are equivalent to (9.3), giving an intrinsic numerical criterion for every admissible transition.

Proof. We use Lemma 9.1. Its full-orthogonal classification and dominance order have exactly the group and self-adjoint convention needed here. The comparison for Grassmannian Segre strata is [4, Theorem 5.1].

The primary spaces $V_i = \ker(C - \alpha_i I)^{m_i}$ are an orthogonal direct sum: the Chinese remainder idempotents are polynomials in the self-adjoint C , hence self-adjoint projections. Each restricted form is nondegenerate, and $C|_{V_i} = \alpha_i I + N_i$, with N_i self-adjoint nilpotent. The prescribed corank is its number of Jordan blocks. The classical theorem gives the indexing. Conversely, a Jordan block is self-adjoint for the reversed-identity Gram matrix; an isometry with the chosen nondegenerate form realizes each partition.

The space of orthogonal decompositions has dimension $(n^2 - \sum_i m_i^2)/2$. Adding the nilpotent orbit dimensions proves (9.2). These are dimensions in the fixed- A matrix space, not in an unmarked moduli stack. Full orthogonal orbits need not be connected; no irreducibility assertion is needed. Theorem 7.3 identifies the unmarked neighbourhood with the pencil, and its spectral module determines the partition at each root. A projective change of the line can therefore do exactly the stated root permutations. For one or two roots the projective stabilizer may be infinite; only its finite permutation image is used.

For necessity, the same polynomial idempotents decompose a family with fixed characteristic polynomial into subbundles of ranks m_i . The rank of N_i^a cannot increase at specialization. Since $m_i - \text{rank } N_i^a = \sum_j \min(a, \lambda_{i,j})$, this gives the length inequalities. Summing the first a columns of conjugate Young diagrams shows that they are equivalent to dominance: conjugation reverses the dominance order.

For sufficiency, fix an orthogonal decomposition of (V, A) . The closure criterion puts a chosen representative of μ_i in the closure of the orbit for λ_i . Thus the special tuple lies in the closure of the product of the generic orbits. Choose an irreducible component of that closure containing the point and meeting the generic orbit product. Successive general hyperplane sections through the point, avoiding the closed complement, give an integral algebraic curve with generic point in that open orbit product. Normalize it and choose a point over the prescribed point. A normal complex curve is smooth. Remove the other points outside the generic orbit product. A constant curve suffices in the equality case. The resulting block operator $\bigoplus_i (\alpha_i I + N_i(t))$ is self-adjoint for the same A . Its determinant is unchanged, and its coranks agree generically and specially, hence throughout the shrunk curve. This gives the required specialization. It does not assert constancy of the full higher Fitting ideals.

For part (3), pull back Theorem 8.1 along the curve's morphism to $P = \text{Gr}(2, \text{Sym}^2 V)$. That theorem identifies the family with the actual ideal-adic neighbourhoods and proves flatness and projectivity under arbitrary base change. No inference from fibrewise isomorphisms is needed. An affine Grassmannian chart around the special pencil supplies a fixed complement to nearby R_C , identifying S_{R_C} , B , and E . There is no relation below degree d , so all lower neighbourhoods become constant. For a strict dominance transition, $\sum_{i,a} (\lambda'_{i,a})^2$ strictly increases; this sum is invariant under permutations of the decorated roots. The generic and special tuples therefore cannot be related by such a permutation. Part (1) separates their unmarked degree- d neighbourhoods. Flatness is asserted only for these finite neighbourhoods, not unrestricted failure schemes. $\square$

Corollary 9.3 (Exactly when the coarse data lose a labelled class). *The number of root-labelled congruence classes with the prescribed data is $\prod_i \#\mathcal{P}(m_i, c_i)$. It is one precisely when, for every i , either $c_i = 1$ or $m_i - c_i \leq 1$. The first possible ambiguity with fixed full determinant and all reduced rank loci occurs in dimension four, at $(m, c) = (4, 2)$, with partitions $(3, 1)$ and $(2, 2)$.*

Proof. Subtracting one from every part identifies $\mathcal{P}(m, c)$ with partitions of $m - c$ into at most c parts. There is one if $c = 1$ or $m - c \leq 1$. Otherwise $(m - c)$ and $(m - c - 1, 1)$ are distinct admissible partitions. The theorem gives the count and the minimal example. $\square$

9.2. The first relation as a standard graded Betti space. The finite invariant has a usual Hilbert-scheme interpretation. The first Betti space means the generator space in the first free module of a minimal graded resolution of the quotient, not the space of relations among generators of its defining ideal. The following statement retains the marked quotient presentation.

Proposition 9.4 (A graded Hilbert locus and its first Betti space). *In the fixed coordinates of Theorem 5.3, let $S = \text{Sym } H^*$, $N = \binom{n+1}{2}$, $M = \binom{N}{2}$, and $Q_R = S/((K_R) + S_{\geq d+1})$. Then*

$$\text{Tor}_1^S(Q_R, \mathbb{C})_d = K_R. \quad (9.5)$$

This identity commutes with arbitrary complex base change. The functor of graded quotients with Hilbert function

$$h_j = \begin{cases} \dim S_j, & 0 \leq j < d, \\ \dim S_d - M, & j = d, \\ 0, & j > d \end{cases}$$

is represented by $\text{Gr}(M, S_d)$. The pencil locus is a closed copy of $\text{Gr}(2, \text{Sym}^2 V)$. Its degree- d first Betti space recovers every scheme-theoretic pencil parameter, including parameters over nonreduced bases.

Proof. Put $J = (K_R) + S_{\geq d+1}$ and $S_+ = \bigoplus_{j>0} S_j$. The augmented sequence $0 \rightarrow J \rightarrow S \rightarrow Q_R \rightarrow 0$ gives $\text{Tor}_1^S(Q_R, \mathbb{C}) = J/S_+J$, since its map to S/S_+ is zero. Its degree- d part is K_R . The same argument over a base algebra applies to the degree- d subbundle with locally free quotient and commutes with pullback. A graded quotient with the indicated locally free components has zero ideal below degree d , a rank- M subbundle in degree d , and the full ideal in higher degrees. Conversely, every such subbundle defines an ideal, as its positive-degree products lie above degree d . This is the Grassmannian functor. The closed immersion and its inverse on the image are Theorem 5.3. This is not an inference from closed-point Torelli to unmarked deformation-groupoid equivalence. $\square$

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